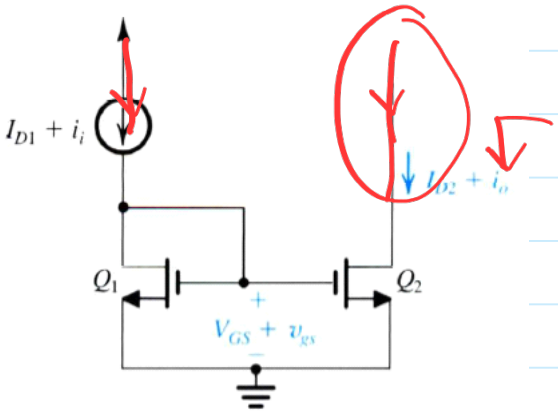
Reading

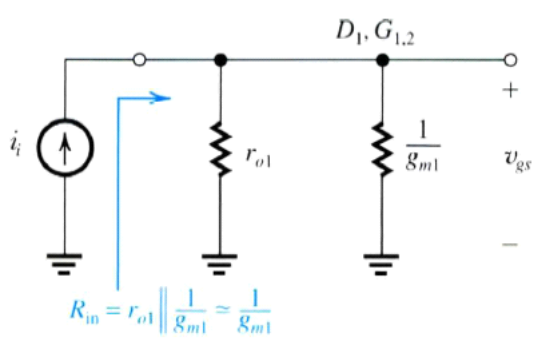
Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

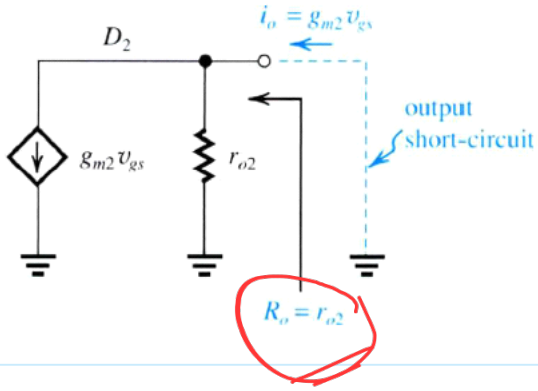
- Chapter 8.4 The Common-Gate and Common-Base Amplifiers as Current Buffers
- Chapter 8.5 The Cascode Amplifier

Recap



~ 50kΩ





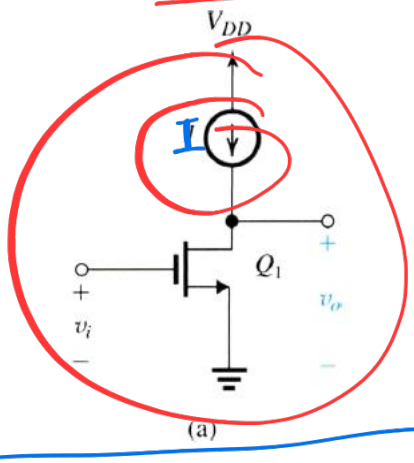
$$R_{in} = r_{o1} \| \frac{1}{g_{m1}} \simeq \frac{1}{g_{m1}}$$

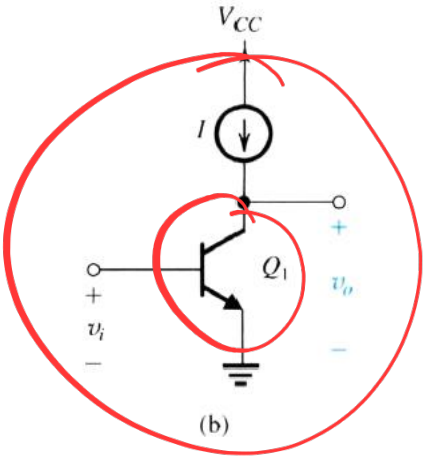
$$R_o = r_{o2}$$

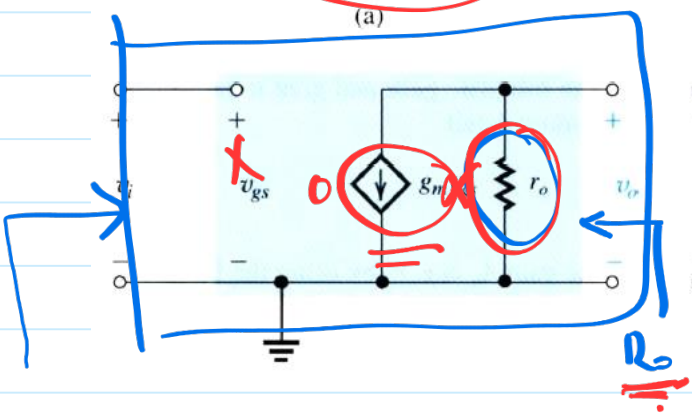
$$A_{is} = \frac{(W/L)_2}{(W/L)_1} \left[\frac{1}{1 + (1/g_{m1} r_{o1})} \right]$$

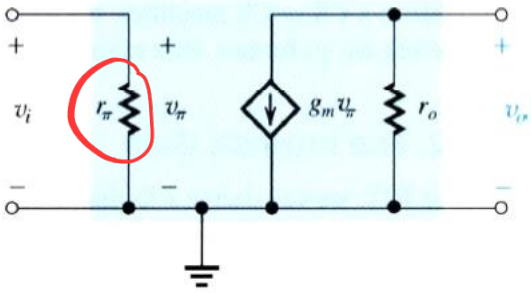
Recap

$A_0 = g_m r_o$









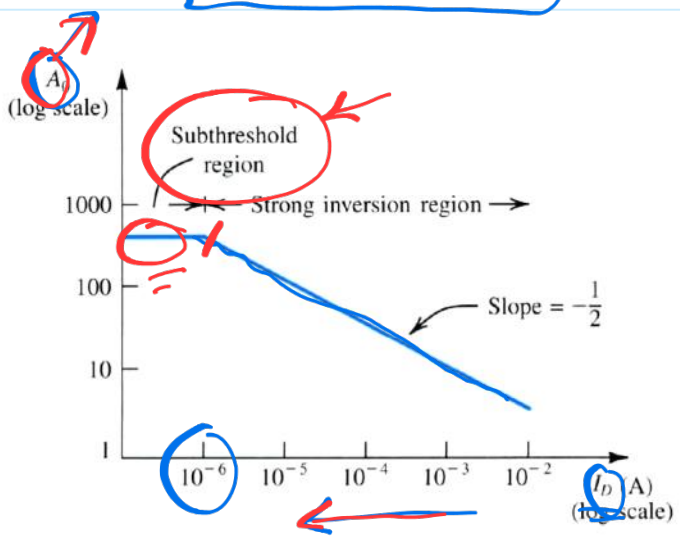
$$R_{in} = \infty$$
$$A_{vo} = -g_m r_o$$
$$R_o = r_o$$

$$R_{in} = r_\pi$$
$$A_{vo} = -g_m r_o$$
$$R_o = r_o$$

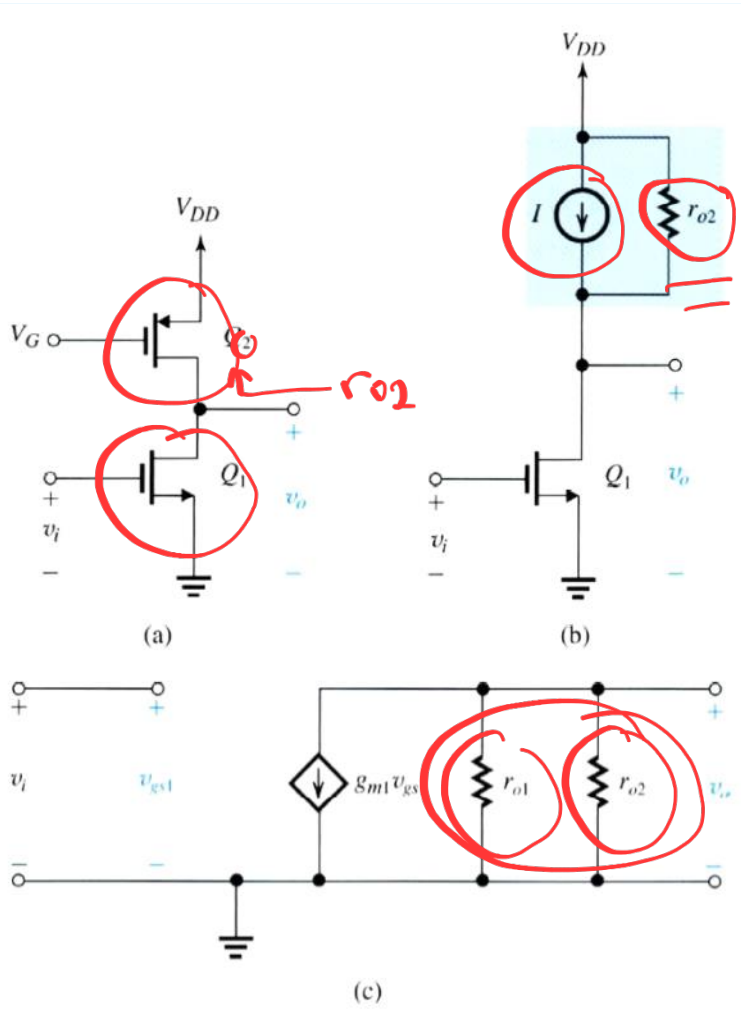
$I = \frac{1}{2}\mu C_{ox}\frac{W}{L}V_{ov}^2$

$$A_0 = \frac{V_A}{V_{OV}/2}$$

$$A_0 = g_m r_o = \frac{V_A}{V_T}$$



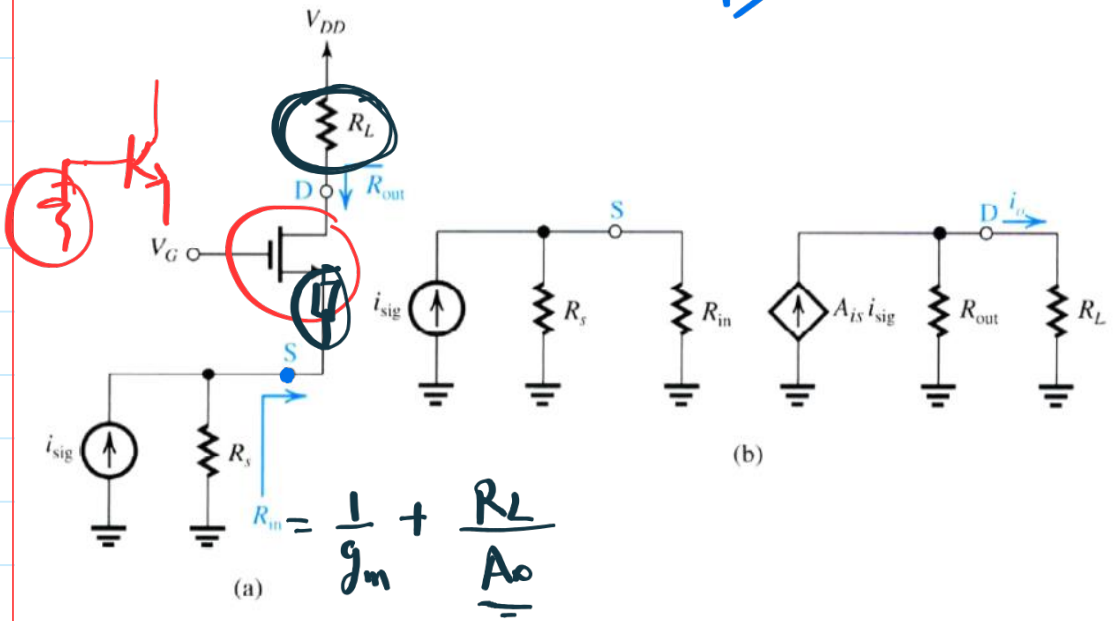
Recap



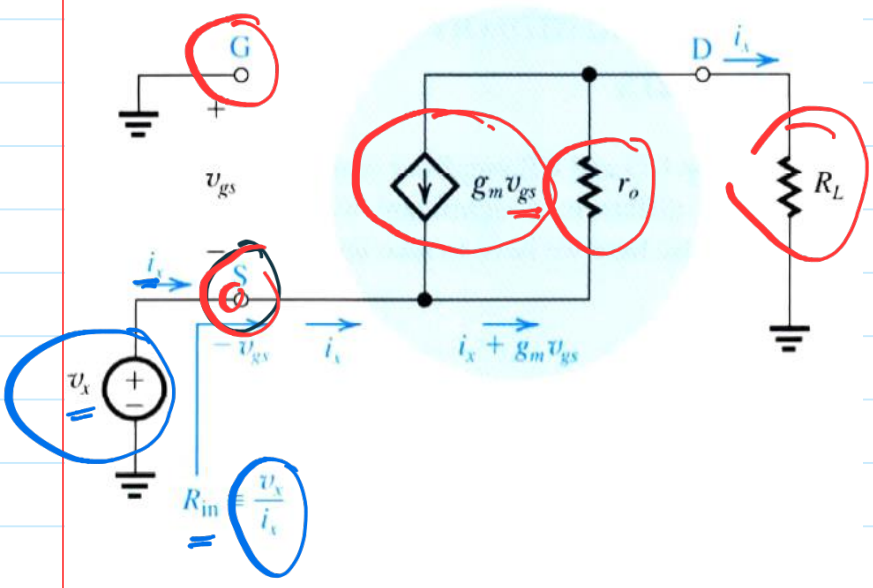
$$A_v \equiv \frac{v_o}{v_i} = -g_{m1}(r_{o1} \| r_{o2}) = -\frac{g_m r_o}{2}$$

Recap

CG



$$R_{in} = \frac{1}{g_m} + \frac{R_L}{A_o}$$

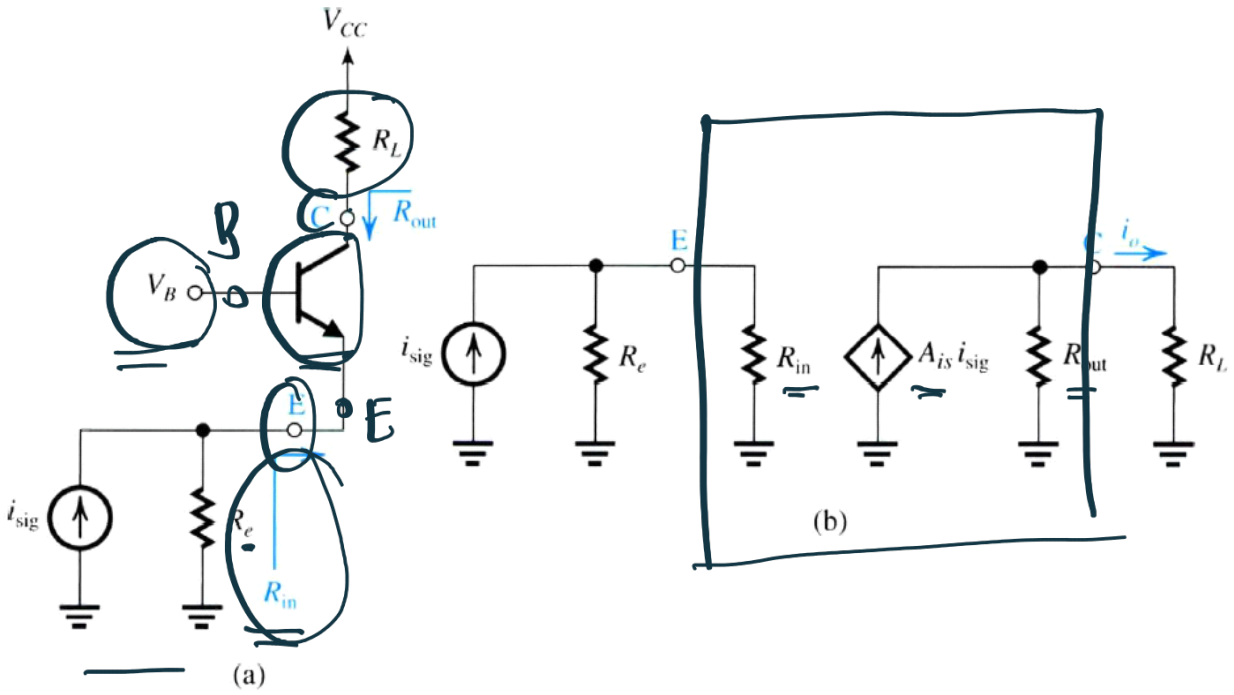


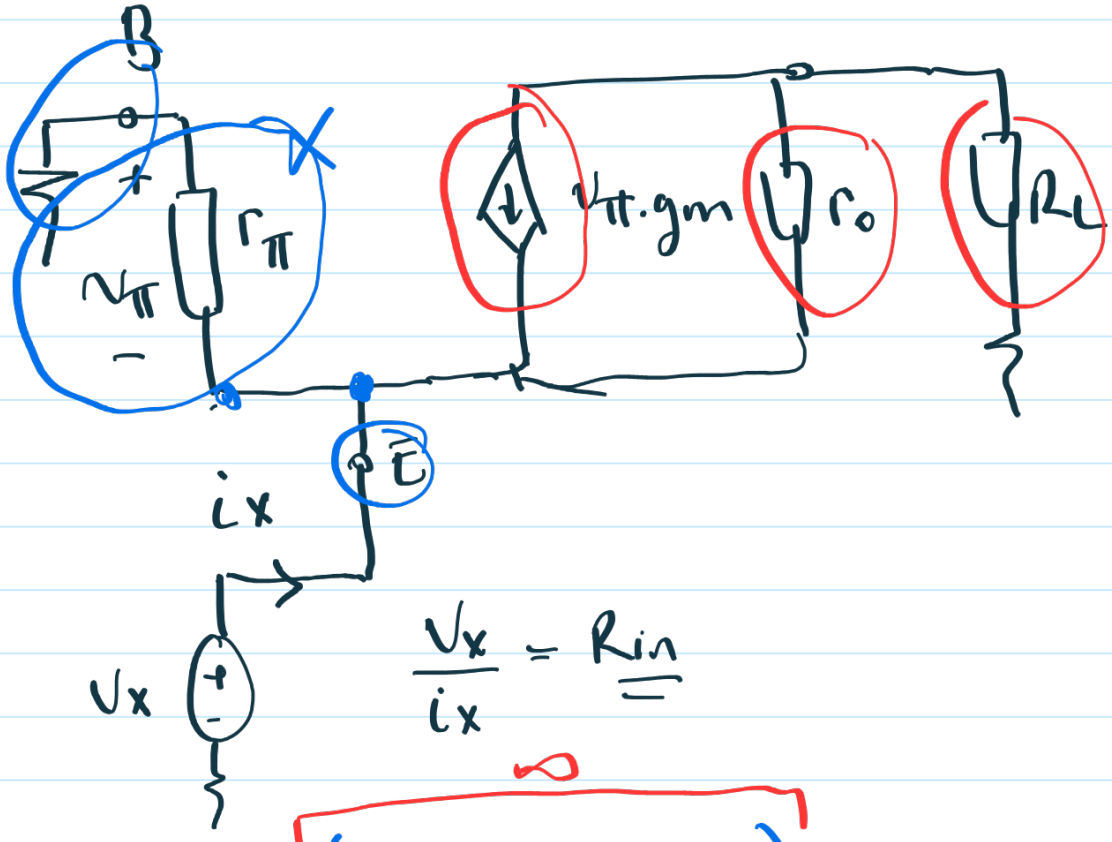
$$R_{in} = \frac{r_o + R_L}{g_m r_o + \cancel{1}}$$

For $g_m r_o \gg 1$,

$$R_{in} \simeq \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

The Rin of the CB Circuit

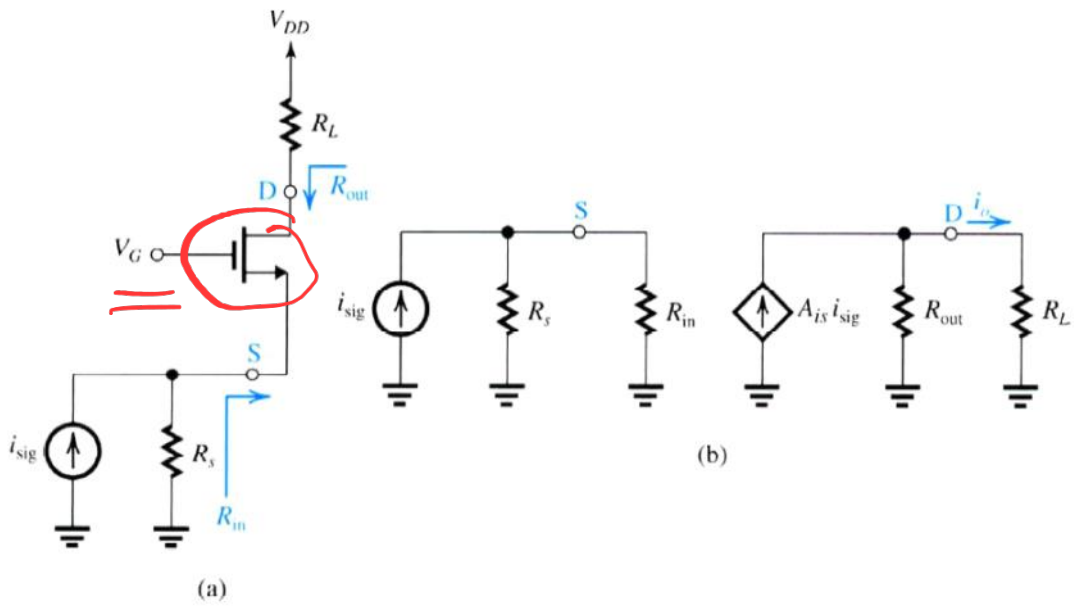


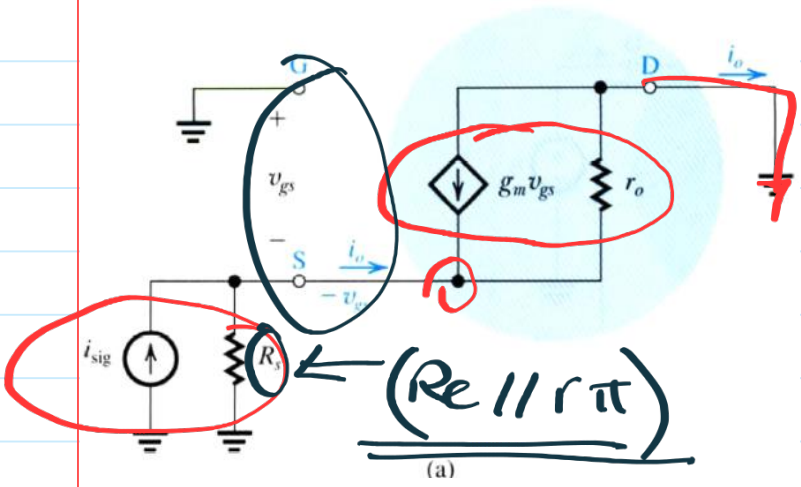


$R_{in\,max} = r_\pi$

$$R_{in} = \left(\frac{1}{g_m} + \frac{R_L}{g_m r_o} \right) \,//\, r_\pi$$

Recap



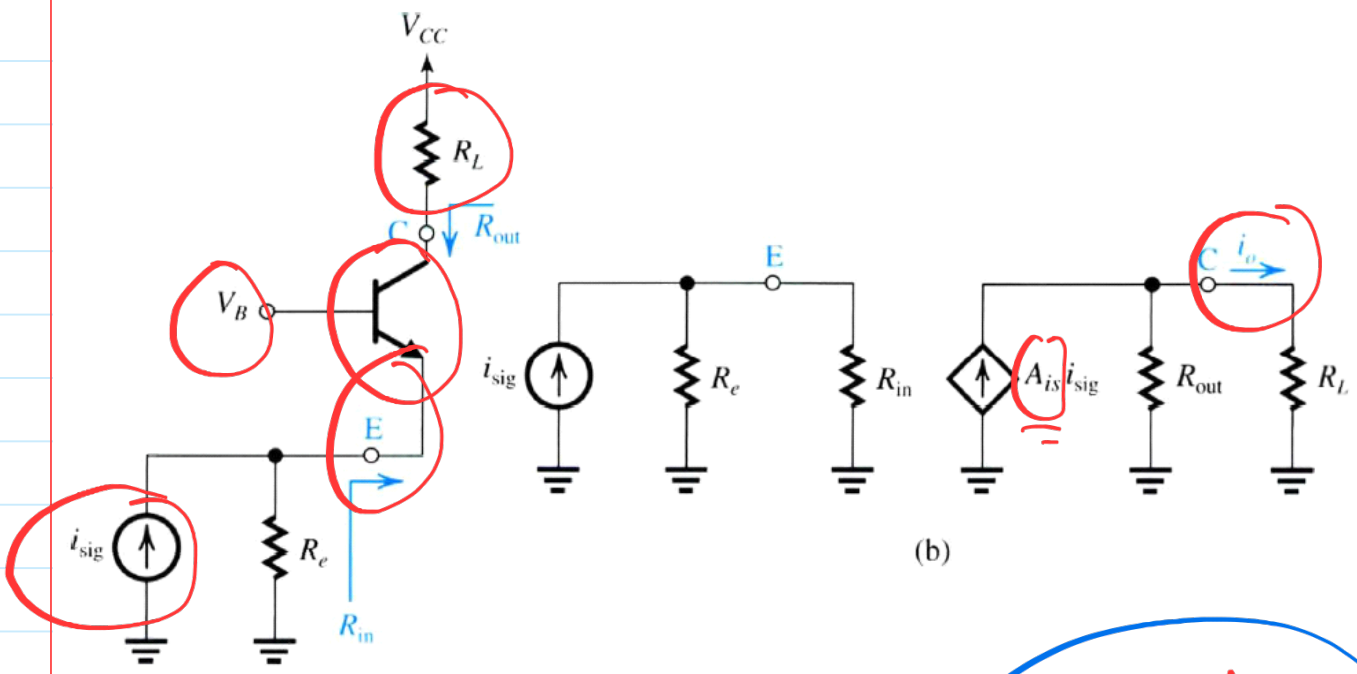


For $g_m r_o \gg 1$ and $g_m R_s \gg 1$,

$$A_{is} = \frac{1 + \frac{1}{g_m r_o}}{1 + \frac{1}{g_m r_o} + \frac{1}{g_m R_s}}$$

$$A_{is} \simeq 1$$

The Short-Circuit Current Gain of CB Circuit



i_o

$g_m v_\pi$ r_o

$\frac{i_o}{i_{sig}} = A_{is}$

i_{sig} $R_e \| r_\pi$

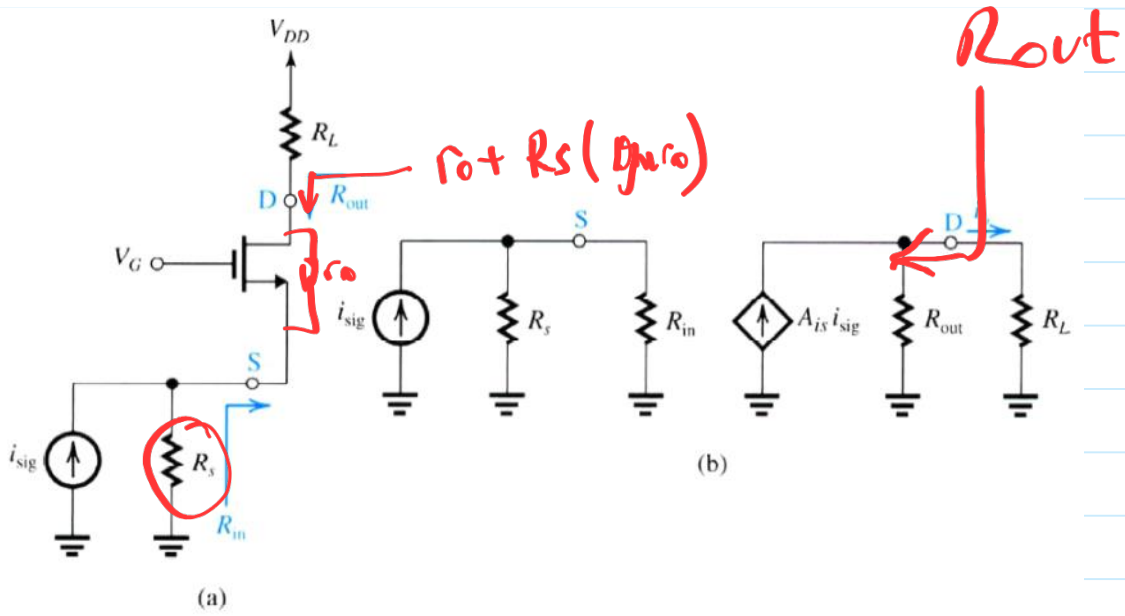
$g_m r_o \gg 1$

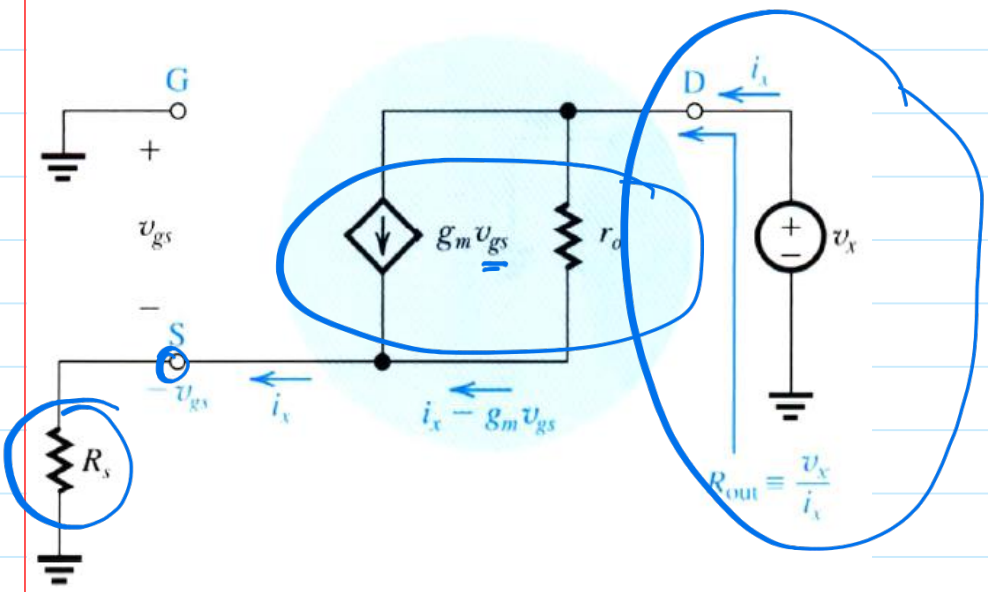
$$A_{is} = \frac{1 + \frac{1}{g_m r_o}}{1 + \frac{1}{g_m r_o} + \frac{1}{g_m (R_e \| r_\pi)}} \quad \frac{1}{\beta}$$

$R_e \gg r_\pi$

$A_{is} \approx 1$

Recap



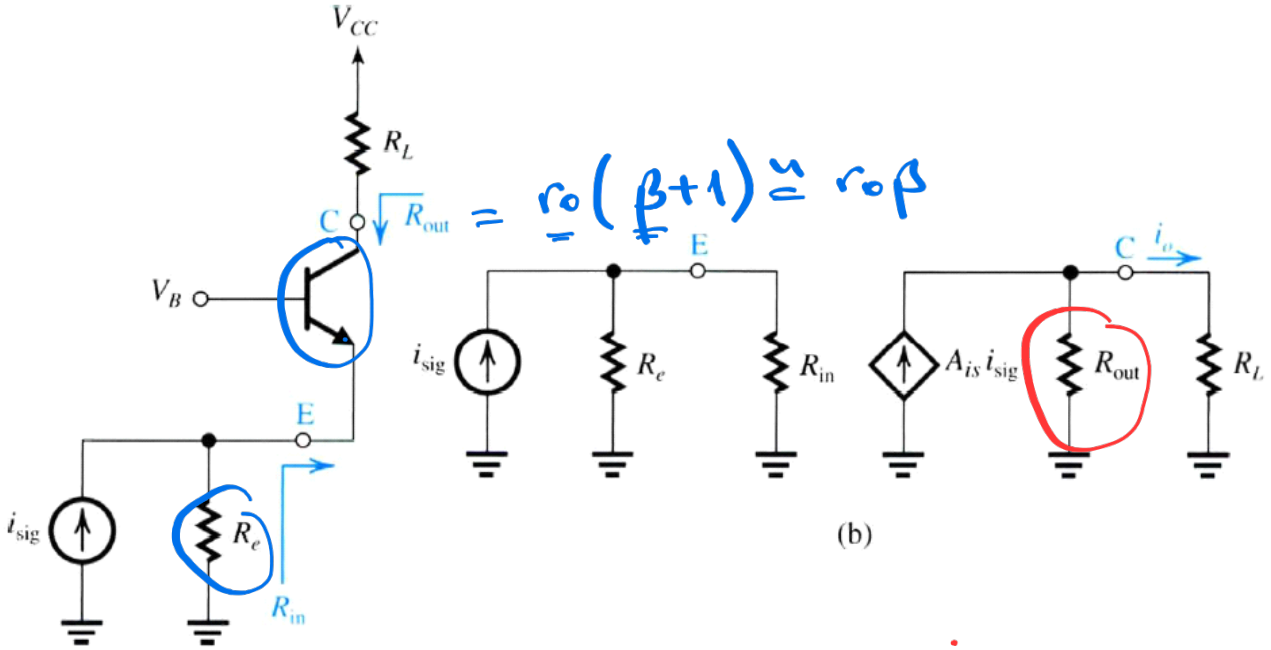


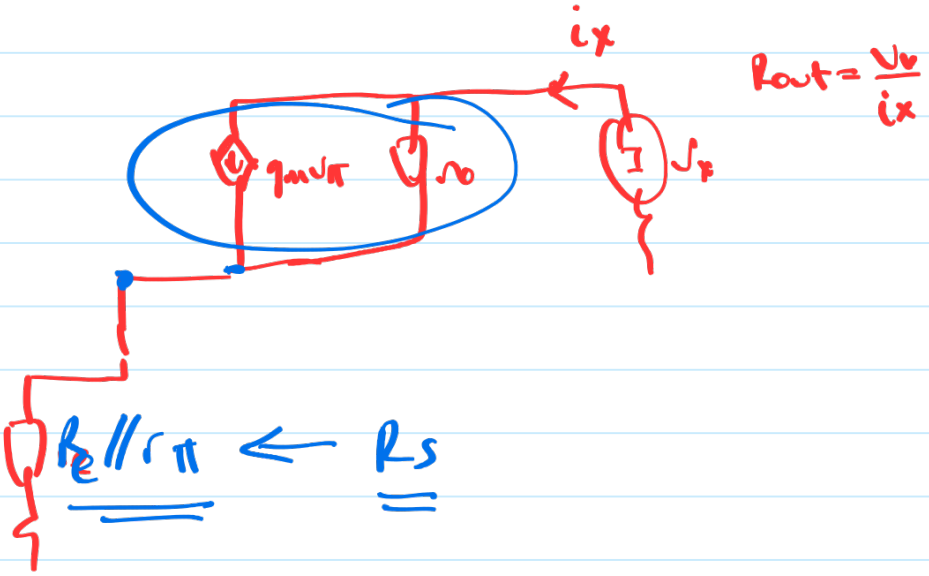
$$R_{out} = r_o + (1 + g_m r_o) R_s$$

For $g_m r_o \gg 1$,

$$R_{out} \simeq r_o + (g_m r_o) R_s$$

The Rout of the CB Circuit





$$R_{out} = r_o + (R_e \| r_\pi) + g_m r_o (R_e \| r_\pi)$$

$$= r_o + (\not{1} + g_m r_o)(R_e \| r_\pi)$$

$$g_m r_o \gg 1$$

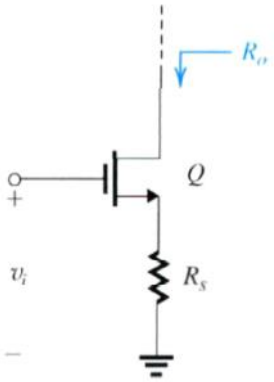
$$R_{out} = r_o + g_m r_o (R_e \| r_\pi) \qquad R_e \to \infty$$

$R_e \to 0$

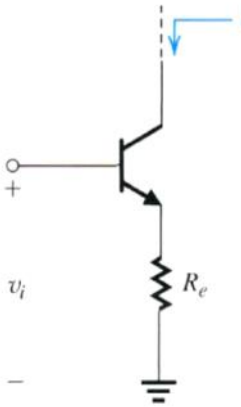
$$\Rightarrow r_o + g_m r_o r_\pi = r_o(1+\beta)$$

($g_m r_\pi = \beta$)

Output Resistance of CS/CE with Rs/Re



$$R_o = R_s + r_o + g_m r_o R_s$$
$$R_o = (1 + g_m R_s)\, r_o$$



$$R_o = r_o + (R_e \| r_\pi) + g_m r_o (R_e \| r_\pi)$$
$$= r_o[1 + g_m (R_e \| r_\pi)]$$

The Cascode Amplifier



Template